

Logarithmic Function

I. Definition

The Napierian (natural) logarithm of x is the function defined by: $f(x) = \ln(x)$, $x \in]0; +\infty[$

- $\ln(1) = 0$
- $\ln(e) = 1$
- $\ln(x) < 0$ when $x \in]0; 1[$
- $\ln(x) > 0$ when $x \in]1; +\infty[$

II. Domain of Definition of $\ln(U(x))$

The function $\ln(U(x))$ is defined for the values of x such that: $U(x) > 0$

Example

Let f , be the function defined by: $f(x) = \ln(2x + 1)$

For $f(x)$ to be defined: $2x + 1 > 0$

$$2x > -1$$

$$x > -\frac{1}{2}$$

Thus, the domain is: $D_f =]-\frac{1}{2}, +\infty[$

Practice

Determine the domain of definition of each of the following functions:

1) $f(x) = \ln(x + 1)$

2) $f(x) = \ln(-x^2 + x + 6)$

3) $f(x) = \ln\left(\frac{2x+1}{x-2}\right)$

4) $f(x) = \frac{1+\ln(x+1)}{2-\ln(x-2)}$

5) $f(x) = \ln[(-x + 3)(2x - 5)]$

Answer:

1) $f(x) = \ln(x + 1)$ is defined when $x + 1 > 0$

$$x > -1$$

$$D_f =] - 1, +\infty[$$

2) $f(x) = \ln(-x^2 - x + 6)$ is defined when $-x^2 - x + 6 > 0$

To find the domain, solve: $-x^2 - x + 6 = 0$

$$x_1 = 2, x_2 = -3$$

Sign table:

x	$-\infty$	-3	2	$+\infty$	
Sign of $-x^2 - x + 6$	$-$	0	$+$	0	$-$

Hence, $D_f =] - 3, 2[$

3) $f(x) = \ln\left(\frac{2x+1}{x-2}\right)$ is defined when $\frac{2x+1}{x-2} > 0$

Solve: $\frac{2x+1}{x-2} = 0$

$$2x + 1 = 0 \text{ and } x - 2 \neq 0$$

$$x = -\frac{1}{2} \text{ and } x \neq 2$$

Sign table:

x	$-\infty$	$-\frac{1}{2}$	2	$+\infty$
$2x + 1$	$-$	0	$+$	$+$
$x - 2$	$-$	$-$	0	$+$
$\frac{2x+1}{x-2}$	$+$	0	$-$	$+$

Thus, the domain is: $D_f =]-\infty, -\frac{1}{2}[\cup]2, +\infty[$

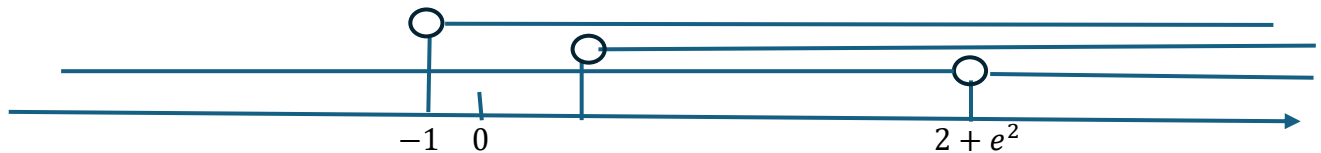
$$4) F(x) = \frac{1+\ln(x+1)}{2-\ln(x-2)}$$

Defined when: $x + 1 > 0$ and $x - 2 > 0$ and $2 - \ln(x - 2) \neq 0$

$$x > -1 \quad \text{and} \quad x > 2 \quad \text{and} \quad \ln(x - 2) \neq 2$$

$$x - 2 \neq e^2$$

$$x \neq 2 + e^2 \approx 9.38$$



Thus, the domain is: $D_f =]2, 2 + e^2[\cup]2 + e^2, +\infty[$

5) $f(x) = \ln [(-x + 3)(2x - 5)]$ is defined when $(-x + 3)(2x - 5) > 0$

Solve each factor: $-x + 3 = 0 \Rightarrow x = 3$

$$2x - 5 = 0 \Rightarrow x = \frac{5}{2}$$

x	$-\infty$	$\frac{5}{2}$	3	$+\infty$
$-x + 3$	+	0	0	-
$2x - 5$	-	0	+	+
$(-x + 3)(2x - 5)$	-	0	+	-

Thus, the domain is: $D_f =]\frac{5}{2}, 3[$

III. Rules of Calculation of $\ln$

(Let $a > 0$ and $b > 0$)

$$1) \ln(a \times b) = \ln(a) + \ln(b)$$

$$2) \ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$$

$$3) \ln(a^r) = r \times \ln(a)$$

$$4) \ln\left(\frac{1}{a}\right) = -\ln(a)$$

$$5) \ln(\sqrt{a}) = \frac{1}{2} \times \ln(a)$$

Examples

$$1) \ln(2 \times 5) = \ln(2) + \ln(5)$$

$$\ln(10) = \ln(2) + \ln(5)$$

$$2) \ln\left(\frac{12}{3}\right) = \ln(12) - \ln(3)$$

$$\ln(4) = \ln(12) - \ln(3)$$

$$3) \ln(2^3) = 3 \times \ln(2)$$

$$\ln(8) = 3 \times \ln(2)$$

$$4) \ln\left(\frac{1}{5}\right) = -\ln(5)$$

$$5) \ln(\sqrt{9}) = \frac{1}{2} \times \ln(9)$$

$$\ln(3) = \frac{1}{2} \times \ln(9)$$

Exercise

Simplify each of the following expressions

$$A = \ln(27) - \ln(9) + \ln(3)$$

$$B = e \times \ln(e) + \ln(\sqrt{e}) - \ln(e^2)$$

Answer:

$$A = \ln(27) - \ln(9) + \ln(3)$$

$$A = \ln\left(\frac{27}{9}\right) + \ln(3)$$

$$A = \ln(3) + \ln(3) = 2 \times \ln(3).$$

$$B = e \times \ln(e) + \ln(\sqrt{e}) - \ln(e^2)$$

$$B = e \times 1 + \frac{1}{2} \times \ln(e) - 2 \times \ln(e)$$

$$B = e + \frac{1}{2} \times 1 - 2 \times 1$$

$$B = e + \frac{1}{2} - 2$$

$$B = \frac{2e + 1 - 4}{2} = \frac{2e - 3}{2}$$

IV. Equations and Inequalities with ln

$$1) \ln(a) = \ln(b) \Leftrightarrow \begin{cases} a > 0 \\ b > 0 \\ a = b \end{cases}$$

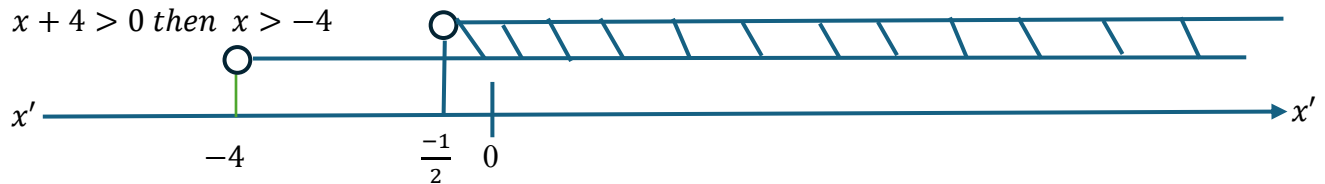
Example

$$\text{Solve: } \ln(2x + 1) = \ln(x + 4)$$

Step 1: Domain conditions

$$2x + 1 > 0 \text{ then } x > -\frac{1}{2}$$

$$x + 4 > 0 \text{ then } x > -4$$



$$\text{Then the domain is } D = \left] -\frac{1}{2}, +\infty \right[$$

Step 2: Equating arguments

$$2x + 1 = x + 4$$

$$2x - x = 4 - 1$$

$$x = 3 \text{ accepted since } 3 \in D$$

$$2) \ln(a) < \ln(b) \Leftrightarrow \begin{cases} a > 0 \\ b > 0 \\ a < b \end{cases}$$

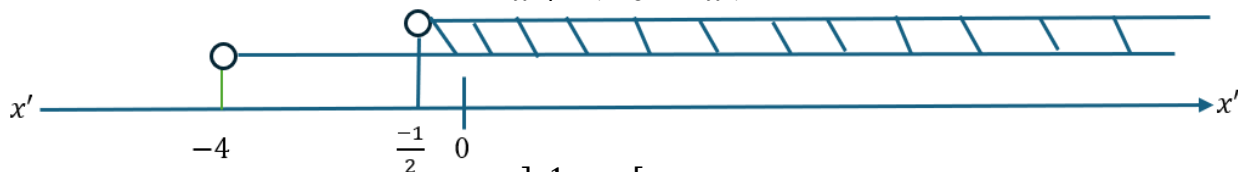
Example

$$\text{Solve: } \ln(2x + 1) < \ln(x + 4)$$

Step 1: Domain conditions

$$2x + 1 > 0 \Rightarrow x > -\frac{1}{2}$$

$$x + 4 > 0 \Rightarrow x > -4$$



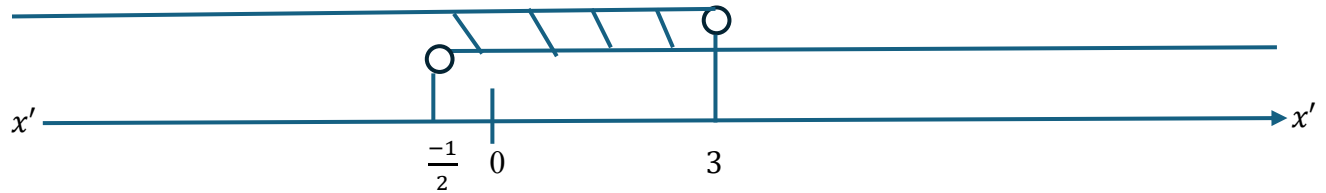
$$\text{Then the domain conditions is } D = \left] -\frac{1}{2}, +\infty \right[$$

Step 2: Inequality of arguments

$$2x + 1 < x + 4 \Rightarrow x < 3$$

Step 3: Intersection of domains

$$x > -\frac{1}{2} \text{ and } x < 3$$



$$\text{Hence, } S =] -\frac{1}{2}, 3[$$

V. Quadratic Equation and Inequality with ln**1) Quadratic Equation with ln .**

$$\text{Solve: } 3(\ln x)^2 + 4 \ln x - 7 = 0$$

$$\text{Condition: } x > 0$$

$$\text{Let } \ln x = A$$

$$\text{Then the equation becomes: } 3A^2 + 4A - 7 = 0$$

$$\text{Solve for } A: \quad A_1 = 1, A_2 = -\frac{7}{3}$$

$$\text{Return to } x: \ln x = 1 \Rightarrow x = e^1$$

$$\ln x = -\frac{7}{3} \Rightarrow x = e^{-\frac{7}{3}}$$

$$\text{Both values satisfy } x > 0 \quad \boxed{x = e \text{ or } x = e^{-\frac{7}{3}}}$$

2) Quadratic Inequality with ln

$$\text{Solve: } 3(\ln x)^2 + 4 \ln x - 7 \leq 0,$$

$$\text{Let } A = \ln x \text{ where } x > 0$$

$$\text{Then the inequality becomes: } 3A^2 + 4A - 7 \leq 0$$

$$\text{Step 2: Solve the quadratic } 3A^2 + 4A - 7 = 0$$

$$A_1 = 1, A_2 = -\frac{7}{3}$$

Return to x $\ln x = 1 \Rightarrow x = e$

$$\ln x = -\frac{7}{3} \Rightarrow x = e^{-\frac{7}{3}}$$

Sign table

x	$-\infty$	$e^{-\frac{7}{3}}$	e	$+\infty$
$3(\ln x)^2 + 4 \ln x - 7$	+	0	-	+

Final Solution $S = [e^{-\frac{7}{3}}, e]$

VI. Derivative of $\ln$

General Rules

$$1) (\ln x)' = \frac{1}{x}$$

$$2) (\ln u)' = \frac{u'}{u}$$

Exercise

Find the derivative function $F'(x)$ of each of the following:

$$1) F(x) = 2 \ln x$$

$$2) F(x) = \ln(3x + 4)$$

$$3) F(x) = \ln \left(\frac{x+1}{2x+3} \right)$$

$$4) F(x) = (\ln x)^2$$

$$5) F(x) = x \ln x$$

$$6) F(x) = \frac{\ln x}{x}$$

$$7) F(x) = \frac{1+\ln x}{x+1}$$

Answer

1) $F(x) = 2 \ln x$

$$F'(x) = 2 \times \frac{1}{x} = \frac{2}{x}$$

2) $F(x) = \ln(3x + 4)$

- Inside function $u = 3x + 4$.

- Derivative: $\frac{u'}{u} = \frac{3}{3x+4}$.

- $F'(x) = \frac{3}{3x+4}$

3) $F(x) = \ln \left(\frac{x+1}{2x+3} \right)$

- Use quotient rule inside t: $\ln u$ with $u = \frac{x+1}{2x+3}$.

- Derivative: $\frac{u'}{u}$.

$$u' = \frac{(1)(2x+3) - (2)(x+1)}{(2x+3)^2} = \frac{2x+3-2x-2}{(2x+3)^2} = \frac{1}{(2x+3)^2}$$

$$F'(x) = \frac{u'}{u} = \frac{\frac{1}{(2x+3)^2}}{\frac{x+1}{2x+3}} = \frac{1}{(x+1)(2x+3)}$$

4) $F(x) = (\ln x)^2$

- Chain rule: derivative of u^2 is $2u \cdot u'$.

- Here $u = \ln x$, so $u' = \frac{1}{x}$.

$$F'(x) = 2 \ln x \times \frac{1}{x} = \frac{2 \ln x}{x}$$

5) $F(x) = x \ln x$

- Product rule: $(uv)' = u'v + uv'$.

- $u = x, u' = 1; v = \ln x, v' = \frac{1}{x}$.

$$F'(x) = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

$$6) F(x) = \frac{\ln x}{x}$$

- Quotient rule: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$.

- $u = \ln x, u' = \frac{1}{x}; v = x, v' = 1$.

$$F'(x) = \frac{\frac{1}{x} \cdot x - (\ln x)(1)}{x^2} = \frac{1 - \ln x}{x^2}$$

$$7) F(x) = \frac{1 + \ln x}{x + 1}$$

- Quotient rule again.

- $u = 1 + \ln x, u' = \frac{1}{x}; v = x + 1, v' = 1$.

$$F'(x) = \frac{\frac{1}{x}(x + 1) - (1 + \ln x)(1)}{(x + 1)^2}$$

$$= \frac{\frac{x + 1}{x} - (1 + \ln x)}{(x + 1)^2}$$

$$= \frac{\frac{x + 1}{x} - 1 - \ln x}{(x + 1)^2}$$

$$= \frac{\frac{x + 1 - x - x \ln x}{x}}{(x + 1)^2} = \frac{\frac{1 - x \ln x}{x}}{(x + 1)^2} = \frac{1 - x \ln x}{x(x + 1)^2}$$

VII. Limits Involving $\ln(x)$

$$1) \lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$2) \lim_{x \rightarrow +\infty} \ln x = +\infty$$

$$3) \lim_{x \rightarrow 0^+} x \ln x = 0^- \quad \textbf{Generalization:} \lim_{x \rightarrow 0^+} x^\alpha \ln x = 0 \text{ for every } \alpha > 0$$

$$4) \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0^+ \quad \textbf{Generalization:} \lim_{x \rightarrow +\infty} \frac{\ln x}{x^\alpha} = 0 \text{ for every } \alpha > 0$$

Examples

$$1) \lim_{x \rightarrow +\infty} (3x \ln x - 4x) = +\infty - \infty \text{ I.F.}$$

$$\lim_{x \rightarrow +\infty} (3x \ln x - 4x) = \lim_{x \rightarrow +\infty} x(3 \ln x - 4) = (+\infty)(+\infty) = +\infty$$

$$2) \lim_{x \rightarrow 0^+} \frac{3 + \ln x}{4 - 2 \ln x} = \frac{+\infty}{-\infty} \text{ I.F.}$$

$$\text{By L'H.R. } \lim_{x \rightarrow 0^+} \frac{3 + \ln x}{4 - 2 \ln x} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{2}{x}} = \frac{-1}{2}$$

$$3) \lim_{x \rightarrow e} \frac{\ln(x) - 1}{x - e} = \frac{\ln e - 1}{e - e} = \frac{1 - 1}{0} = \frac{0}{0} \text{ I.F.}$$

$$\text{By L'H.R. } \lim_{x \rightarrow e} \frac{\ln(x) - 1}{x - e} = \lim_{x \rightarrow e} \frac{\frac{1}{x}}{1} = \frac{1}{e}$$

$$4) \lim_{x \rightarrow +\infty} (x^2 - \ln x) = +\infty - \infty \text{ I.F.}$$

$$\lim_{x \rightarrow +\infty} (x^2 - \ln x) = \lim_{x \rightarrow +\infty} x^2 \left(1 - \frac{\ln x}{x^2}\right) = (+\infty)(1 - 0) = +\infty \quad \text{Since } \frac{\ln x}{x^2} \rightarrow 0,$$

$$5) \lim_{x \rightarrow +\infty} \frac{x + 1 - \ln x}{x^2} = \lim_{x \rightarrow +\infty} \left(\frac{x}{x^2} + \frac{1}{x^2} - \frac{\ln x}{x^2}\right) = \lim_{x \rightarrow +\infty} \left(\frac{1}{x} + \frac{1}{x^2} - \frac{\ln x}{x^2}\right) = 0$$

VIII. Variations of the Logarithmic Function

We study the function: $f(x) = \ln(x)$, $x > 0$

Limits and Asymptotes

$\lim_{x \rightarrow 0^+} \ln(x) = -\infty$ Thus, the line $x = 0$ is a **vertical asymptote**.

$$\lim_{x \rightarrow +\infty} \ln(x) = +\infty$$

Derivative and Monotonicity

$$f'(x) = \frac{1}{x} > 0 \text{ for all } x > 0$$

Therefore, $f(x)$ is **strictly increasing** on its domain

Particular Points

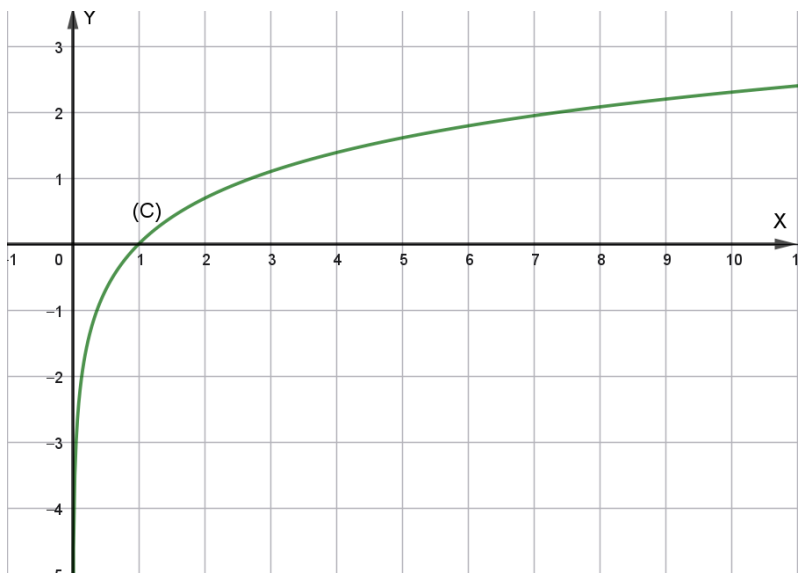
$$f(1) = 0 \text{ and } f(e) = 1$$

So, the representative curve (C) of f passing through the points $(1, 0)$ and $(e, 1)$

Table of variations

x	0	$+\infty$
$f'(x)$	N	+
$f(x)$	$-\infty$	$+\infty$

Representative curve (C) of the function f defined by: $f(x) = \ln(x)$



IX. Primitives

$$1) \int \frac{1}{x} dx = \ln |x| + C$$

$$2) \int \frac{u'}{u} dx = \ln |u| + C$$

Examples

$$1) \int \frac{2x+1}{x^2+x+1} dx$$

$$\text{Let } u = x^2 + x + 1. \quad u' = 2x + 1,$$

$$\text{Thus } \int \frac{2x+1}{x^2+x+1} dx = \int \frac{u'}{u} dx = \ln |u| + C = \ln |x^2 + x + 1| + C$$

$$2) \int \frac{3x^2+2}{2x^3+4x} dx$$

$$\text{Let } u = 2x^3 + 4x \quad u' = 6x^2 + 4.$$

$$\int \frac{3x^2 + 2}{2x^3 + 4x} dx = \frac{1}{2} \int \frac{6x^2 + 4}{2x^3 + 4x} dx = \frac{1}{2} \int \frac{u'}{u} dx = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |2x^3 + 4x| + C$$